D.K.T.E. Society's Textile and Engineering Institute, Ichalkaranji

Project Title: Water Purification Awareness

PROJECT TEAM:

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Introduction

Clean drinking water is essential for good health. Water contamination and improper storage can increase the risk of waterborne diseases. Our project focused on water purification awareness, hygiene, water conservation, and plant-based treatment approaches.

Problem Statement

- Lack of awareness about safe drinking water and storage.
- Risk of water contamination due to poor hygiene and sanitation.
- Need for better water conservation and environmental cleanliness.

Objectives

- Create awareness about safe drinking water.
- Educate students about hygiene and contamination prevention.
- Study plant-based water treatment approaches.
- Understand municipal water supply and sanitation.

Project Approach

- Selected a water body of approximately 10 × 10 metres for the study.
- Selected *Canna indica* and *Typha angustifolia* to explore their potential role in plant-based water treatment.
- Visited a chemical laboratory to learn about water purification.
- Conducted a school awareness visit and a Mahanagarpalika visit.
- Documented activities using photographs and notes.

Materials and Concepts Studied

- Selected aquatic plants.
- Coconut coir or coconut bark with soil as a proposed plant-support method.
- Laboratory test tubes and a water purification awareness chart.

Week 1 – Project Selection
Selected Water Purification Awareness as our Community Engagement Project and discussed the importance of clean drinking water.

Week 2 – Project Planning
Finalized the project objectives and planned activities related to water awareness, hygiene, and conservation.

Week 3 – Selection of Water Body and Plants
Selected a water body of approximately 10 × 10 metres and selected *Canna indica* and *Typha angustifolia* for studying the plant-based approach.

Week 4 – Chemical Laboratory Visit
Visited the chemical laboratory, observed the water purification awareness chart and test tubes, and captured photographs for documentation.

Week 5 – School Awareness Visit
Visited a school and shared information about safe drinking water, proper water storage, handwashing, and prevention of water contamination.

Week 6 – Mahanagarpalika Visit
Learned about municipal water supply, drainage management, sanitation, and public cleanliness.

Week 7 – Plant-Based Treatment Approach
Studied the potential use of *Canna indica* and *Typha angustifolia*. Explored coconut coir or coconut bark with soil as a possible support for plants near the water surface.

Week 8 – Documentation and Review
Organized the information and photographs collected during the visits and prepared the project report and presentation.

Chemical Laboratory Visit

- Observed a water purification awareness chart.
- Observed laboratory test tubes and related materials.
- Learned about the importance of water purification and testing.
- Captured photographs as project evidence.

Plant-Based Approach

- Studied *Canna indica* and *Typha angustifolia*.
- Explored the proposed use of coconut coir or coconut bark with soil to support plants near the water surface.
- Identified the need for continued observation and laboratory water-quality testing.

School Awareness Visit

- Explained the importance of safe drinking water.
- Shared information about proper water storage and handwashing.
- Discussed prevention of water contamination and water conservation.
- Promoted cleanliness and hygiene among students.

Mahanagarpalika Visit

- Learned about municipal water supply.
- Understood drainage and sanitation management.
- Observed the importance of public cleanliness and community health.

Learning: These visits helped us understand the importance of awareness and local authorities in maintaining public health.

Work Completed

- Selected the project topic, study area, and plants.
- Completed the chemical laboratory visit.
- Conducted a school awareness visit.
- Conducted a Mahanagarpalika visit.
- Documented the activities and prepared the project report.

Learning Outcomes

- Improved understanding of water safety, hygiene, and sanitation.
- Learned about municipal water management.
- Developed teamwork, communication, and documentation skills.

Future Scope

- Continue observing the proposed plant-based approach.
- Conduct laboratory tests to assess water quality.
- Extend water-safety awareness to more students and community members.

Conclusion

- The Water Purification Awareness project helped us understand the importance of clean drinking water, hygiene, sanitation, and water conservation.
- Through the laboratory visit, school awareness campaign, and Mahanagarpalika visit, we gained practical knowledge about water purification and community water management.
- We also studied a plant-based treatment approach and identified the need for further observation and laboratory testing before drawing conclusions about water quality.

THANK YOU…

"Every Drop Counts — Keep Water Clean and Safe."